pargo

PARGO PARCELS

LAST-MILE LOGISTICS INTELLIGENCE

Data Warehouse Portfolio

End-to-end analytics platform covering data engineering, dimensional modelling, machine learning, and business intelligence for South Africa's leading parcel pickup network.

30.2M	152M+	36	15
Parcels	Tracking Events	Months	ML Models

Technology Stack: Snowflake DW | PostgreSQL | DBT | Python | XGBoost | LightGBM | Snowpark ML

Coverage: South Africa | 9 Provinces | July 2023 – June 2026

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Table of Contents

1. Project Overview & Business Context

About Pargo

Pargo is South Africa's leading parcel pickup network, enabling e-commerce retailers and consumers to send, receive, and return parcels through a nationwide network of over 4,000 pickup points (PUDOs) located in convenience stores, petrol stations, and retail outlets. Operating across all nine provinces, Pargo bridges the gap between online retail and the practical realities of last-mile delivery in South Africa, where home delivery is often unreliable or uneconomical.

This portfolio demonstrates a production-grade data warehouse platform built to power operational analytics, business intelligence, and machine learning for Pargo's logistics network. The platform ingests 36 months of synthetic but structurally authentic parcel lifecycle data, transforming raw events into curated analytical models.

Business Objectives

Operational Visibility	Real-time and historical insight into parcel throughput, dwell times, SLA compliance, and exception rates across all pickup points and couriers.
RTS Risk Reduction	Predict which parcels are likely to be Returned to Sender before it happens, enabling proactive intervention to improve delivery success rates.
Customer Intelligence	Understand customer lifetime value, segment customers by behaviour, and identify churn risk before customers disengage.
Demand Forecasting	Forecast parcel volumes 1-6 months ahead to support staffing, capacity planning, and courier contract negotiations.
Courier Performance	Score and rank courier agents and logistics partners on reliability, speed, and exception rates to drive SLA accountability.
Retailer Scorecard	Provide retail partners with performance benchmarks covering dispatch quality, packaging compliance, and return rates.

Data Scope

30.2M	152.8 M	10M	36 mths	4,000 +	150
Total Parcels	Tracking Events	Customers	Data Window	Pickup Points	Retail Partners

The dataset spans July 2023 to June 2026 and covers the full parcel lifecycle: order creation, dispatch, in-transit scan events, pickup point arrival, customer collection, and returns. Each province is represented in proportion to South Africa's population distribution, with Gauteng (35%), Western Cape (20%), and KwaZulu-Natal (18%) forming the majority of parcel volume.

Technical Architecture Summary

Ingestion	Python bulk loader using Snowflake PUT + COPY INTO with parquet files organised by year/month partitions. Parallel batch upload achieves ~1M rows/min.

Warehouse	Snowflake cloud data warehouse (af-south-1, AWS) with RAW / STAGING / MARTS / ML_FEATURES schemas. 8 tables, 182M+ rows total.
Transform	DBT manages all SQL transformations from raw tables through staging views to analytical mart tables and ML feature tables.
ML / AI	Python scikit-learn, XGBoost, LightGBM, and Snowpark ML for 15 production machine learning models across 5 problem domains.
BI Layer	HTML5 + Chart.js interactive dashboard with 6 analytical tabs. ML Visuals portal with 33 charts. Portfolio ebook PDF.
PostgreSQL	All mart and ML feature table SQL is ANSI-compatible. Full Snowflake → PostgreSQL migration guide included in Section 8.

2. Data Architecture & Entity Relationship Diagram

The warehouse follows a classic star schema dimensional model. A star schema places business process facts (measurable events) at the centre, surrounded by descriptive dimension tables. This design is optimised for analytical query performance, intuitive navigation, and compatibility with BI tools.

Schema Design Principles

- Fact tables store transactional records with foreign keys to dimensions and numeric measures.
- Dimension tables store descriptive attributes that provide context to facts.
- Surrogate integer keys are used throughout for join performance.
- All fact tables include LOAD_YEAR and LOAD_MONTH for partition-aware queries.
- Timestamps stored as TIMESTAMP_NTZ (no timezone) for consistent cross-region comparison.
- GPS coordinates (latitude/longitude) captured on tracking events for spatial analysis.

Entity Relationship Overview

From Table	Card.		To Table	Relationship
DIM_CUSTOMERS	1	---<	FACT_ORDERS	Customer places orders
DIM_RETAILERS	1	---<	FACT_ORDERS	Retailer initiates orders
FACT_ORDERS	1	---<	FACT_PARCELS	Order contains parcels
DIM_COURIERS	1	---<	FACT_PARCELS	Courier assigned to parcel
DIM_PICKUP_POINTS	1	---<	FACT_PARCELS	Parcel delivered to PUDO
FACT_PARCELS	1	---<	FACT_TRACKING_EVENTS	Parcel has scan events
FACT_PARCELS	1	---<	FACT_RETURNS	Parcel may be returned
DIM_RETAILERS	1	---<	FACT_RETURNS	Retailer receives return

Note: DIM_PICKUP_POINTS also links to FACT_RETURNS via DROP_OFF_POINT_ID, capturing where a return was dropped off.

Data Flow Layers

Layer 1 — Raw	Parquet files loaded via PUT + COPY INTO into Snowflake RAW schema. LOAD_YEAR and LOAD_MONTH partition columns added at load time.
Layer 2 — Staging	DBT views (stg_*) clean, standardise column names, cast data types, and handle nulls. Views run on-demand — no storage cost.
Layer 3 — Marts	DBT materialised tables (mart_*) aggregate staging data into analytical models. Primary source for dashboards and reports.
Layer 4 — ML Features	DBT feature tables (feat_*) produce wide, denormalised feature sets ready for Python ML training pipelines.

3. Dimensional Model: Tables & Schema

The warehouse contains 4 dimension tables and 4 fact tables across the RAW schema. Below is the full column-level definition for each table.

DIM_CUSTOMERS

One record per customer. Province and registration date enable cohort and geographic analysis.

Column	Type	Description
CUSTOMER_ID	NUMBER(38)	Surrogate primary key
CUSTOMER_NAME	VARCHAR	Full name (anonymised in production)
EMAIL	VARCHAR	Contact email
PHONE	VARCHAR	Mobile number
PROVINCE	VARCHAR	SA province (9 values)
REGISTRATION_DATE	DATE	Date the customer first registered
CUSTOMER_SEGMENT	VARCHAR	New / Active / Loyal / VIP
ACTIVE_FLAG	NUMBER(1)	1 = active last 90 days

DIM_RETAILERS

150 retail partners. Retailer scorecard mart is built on top of this dimension.

Column	Type	Description
RETAILER_ID	NUMBER(38)	Surrogate primary key
RETAILER_NAME	VARCHAR	Trading name
CATEGORY	VARCHAR	Fashion / Electronics / Health / Home / Sports
CONTACT_EMAIL	VARCHAR	Account manager email
ONBOARDING_DATE	DATE	Date retailer joined Pargo
ACTIVE	BOOLEAN	Whether retailer is currently active

DIM_COURIERS

500 courier agents contracted by Pargo across all provinces.

Column	Type	Description
COURIER_ID	NUMBER(38)	Surrogate primary key
COURIER_NAME	VARCHAR	Courier or fleet company name
VEHICLE_TYPE	VARCHAR	Motorbike / Car / Van / Truck
PROVINCE	VARCHAR	Primary operating province
RATING	FLOAT	Performance rating (1–5 scale)

ACTIVE	BOOLEAN	Whether courier is currently active

DIM_PICKUP_POINTS

4,000+ PUDO (Pickup and Drop-Off) points. Province and GPS coordinates enable spatial analysis and service area coverage mapping.

Column	Type	Description
PICKUP_POINT_ID	NUMBER(38)	Surrogate primary key
POINT_NAME	VARCHAR	Trading name of the pickup point
ADDRESS	VARCHAR	Street address
CITY	VARCHAR	City
PROVINCE	VARCHAR	SA province
LATITUDE	FLOAT	GPS latitude
LONGITUDE	FLOAT	GPS longitude
ACTIVE	BOOLEAN	Operational status

FACT_PARCELS

The central fact table with 30.2M rows. One record per parcel. Tracks the full lifecycle from dispatch to final status. RTS risk prediction targets this table.

Column	Type	Description
PARCEL_ID	NUMBER(38)	Surrogate primary key
ORDER_ID	NUMBER(38)	FK to FACT_ORDERS
RETAILER_ID	NUMBER(38)	FK to DIM_RETAILERS
COURIER_ID	NUMBER(38)	FK to DIM_COURIERS
PICKUP_POINT_ID	NUMBER(38)	FK to DIM_PICKUP_POINTS
PARCEL_STATUS	VARCHAR	Dispatched / In-Transit / At PUDO / Collected / RTS / Damaged
PARCEL_WEIGHT_KG	FLOAT	Gross weight in kilograms
PARCEL_VALUE_ZAR	FLOAT	Declared value in South African Rand
DELIVERY_COST_ZAR	FLOAT	Delivery cost per parcel
DISPATCHED_ZAR	TIMESTAMP_NTZ	When retailer dispatched from warehouse
ARRIVED_AT_PUDO	TIMESTAMP_NTZ	When parcel arrived at pickup point
COLLECTED_AT	TIMESTAMP_NTZ	When customer collected parcel
DWELL_DAYS	NUMBER	Days parcel sat at the PUDO
SLA_FLAG	NUMBER(1)	1 = within SLA, 0 = breach
LOAD_YEAR	NUMBER	Partition year
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM format)

FACT_TRACKING_EVENTS

152.8M rows — the largest table. Each row is a scan or status change event emitted by courier scanners, mobile apps, or PUDO terminals. Includes GPS coordinates for heatmap and spatial analytics.

Column	Type	Description
TRACKING_EVENT_ID	NUMBER(38)	Surrogate primary key
PARCEL_ID	NUMBER(38)	FK to FACT_PARCELS
EVENT_TYPE	VARCHAR	CollectedByDriver / ArrivedAtHub / ArrivedAtPUDO / CustomerPickup / ReturnInitiated
EVENT_TIMESTAMP	TIMESTAMP_NTZ	When the event occurred
LATITUDE	FLOAT	GPS latitude at time of event
LONGITUDE	FLOAT	GPS longitude at time of event
STATUS	VARCHAR	Parcel status at time of event

SOURCE_SYSTEM	VARCHAR	Mobile App / PUDO Terminal / Courier Scanner
LOAD_YEAR	NUMBER	Partition year
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM)

FACT_RETURNS

Captures parcel return transactions. Return reason codes and monetary values support returns analytics and retailer chargebacks.

Column	Type	Description
RETURN_ID	NUMBER(38)	Surrogate primary key
PARCEL_ID	NUMBER(38)	FK to FACT_PARCELS
RETAILER_ID	NUMBER(38)	FK to DIM_RETAILERS
RETURN_INITIATED_AT	TIMESTAMP_NTZ	When return was triggered
RETURN_REASON	VARCHAR	CustomerNotAvailable / WrongAddress / Damaged / Refused / Expired
RETURN_VALUE_ZAR	FLOAT	Value of returned goods
DROP_OFF_POINT_ID	NUMBER(38)	FK to DIM_PICKUP_POINTS
LOAD_YEAR	NUMBER	Partition year
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM)

4. DBT Transformation Layer

DBT (Data Build Tool) manages all SQL transformations from raw tables through staging views to analytical mart tables and ML feature tables. DBT enforces data quality tests, generates documentation, and maintains a full lineage DAG.

Staging Views (stg_*)

8 staging views sit directly on top of the RAW schema tables. They perform standardisation, type casting, and null handling. Views are not materialised — they execute on-demand with no storage cost.

stg_customers	Standardises province names, casts registration_date, filters deleted records.
stg_retailers	Normalises category labels, joins retailer type lookup.
stg_couriers	Adds vehicle_class derived column from vehicle_type.
stg_pickup_points	Validates GPS coordinates, adds province_code lookup.
stg_parcels	Computes dwell_days = DATEDIFF(collected_at, arrived_at_pudo), adds rts_flag.
stg_tracking_events	Filters duplicate scans, adds event_sequence_number per parcel.
stg_returns	Joins to stg_parcels for enriched return context.
stg_orders	Aggregates order-level metrics from parcel records.

Mart Tables (mart_*)

4 materialised tables serve as the primary analytical layer consumed by dashboards, reports, and ad-hoc queries.

mart_parcel_summary	One row per parcel with all dimension attributes denormalised. Includes RTS flag, dwell days, SLA flag, delivery cost. ~30M rows. Primary table for operational reporting.
mart_retailer_scorecard	One row per retailer per month. Aggregates dispatch volume, RTS rate, SLA rate, avg dwell time, and packaging exception rate. Powers the Retailer Analytics tab.
mart_pudo_performance	One row per pickup point per month. Captures throughput, avg dwell time, collection rate, and exception count. Used for PUDO capacity planning.
mart_customer_activity	One row per customer per month. Tracks order count, parcel volume, RTS rate, and last active date. Powers CLV and churn models.

ML Feature Tables (feat_*)

3 wide, denormalised feature tables are materialised specifically for Python ML training pipelines. They join mart tables and add engineered features.

feat_rts_model	One row per parcel for RTS binary classification. 14 features including weight, value, dwell, exception_count, transit_hours, province, vehicle_type, retailer_category. Target: rts_flag (1/0).

feat_customer_ltv	One row per customer. Features: tenure_days, order_frequency, avg_order_value, rts_rate, engagement_score, province. Used for CLV regression and churn classification.
feat_volume_forecast	Monthly aggregated volume by province. Features: month_index, month_of_year, is_peak_season, rolling_3m_avg. Used for Holt-Winters and LightGBM forecasting.

DBT Tests

DBT schema tests are defined for all models to ensure data quality:

- not_null tests on all primary keys and critical foreign keys
- unique tests on PARCEL_ID, CUSTOMER_ID, RETAILER_ID, COURIER_ID
- accepted_values tests on PARCEL_STATUS, RETURN_REASON, CUSTOMER_SEGMENT
- relationships tests verifying FK integrity between fact and dimension tables
- custom test: rts_rate_sanity — fails if overall RTS rate exceeds 40%

5. Machine Learning Models

Building this ML layer required translating raw operational data into predictive intelligence. I designed, engineered, and evaluated 15 models spanning 5 problem domains — selecting algorithms based on the specific nature of each business question, handling class imbalance, engineering features from 152M tracking events, and packaging results into Snowflake-native stored procedures for production deployment.

Python	Scikit-learn	XGBoost	LightGBM	Statsmodels	Snowpark ML	Feature Engineering	Model Evaluation

15	0.89	77.8%	R50	5%
Models Trained	Best ROC-AUC	Forecast Fit R^2	CLV MAE	Anomaly Rate

WHY THIS ANALYSIS

A single dashboard summarising all 15 models was built to give stakeholders a one-page view of the full ML portfolio — covering classifier performance, forecast accuracy, and segmentation quality side-by-side. This reflects my approach of not just building models but communicating results clearly to non-technical audiences.

KEY FINDING

Stacking Ensemble leads with ROC-AUC 0.89. The CLV regressor achieves R^2 0.987 with MAE of just R50 — near-perfect prediction of customer revenue potential.

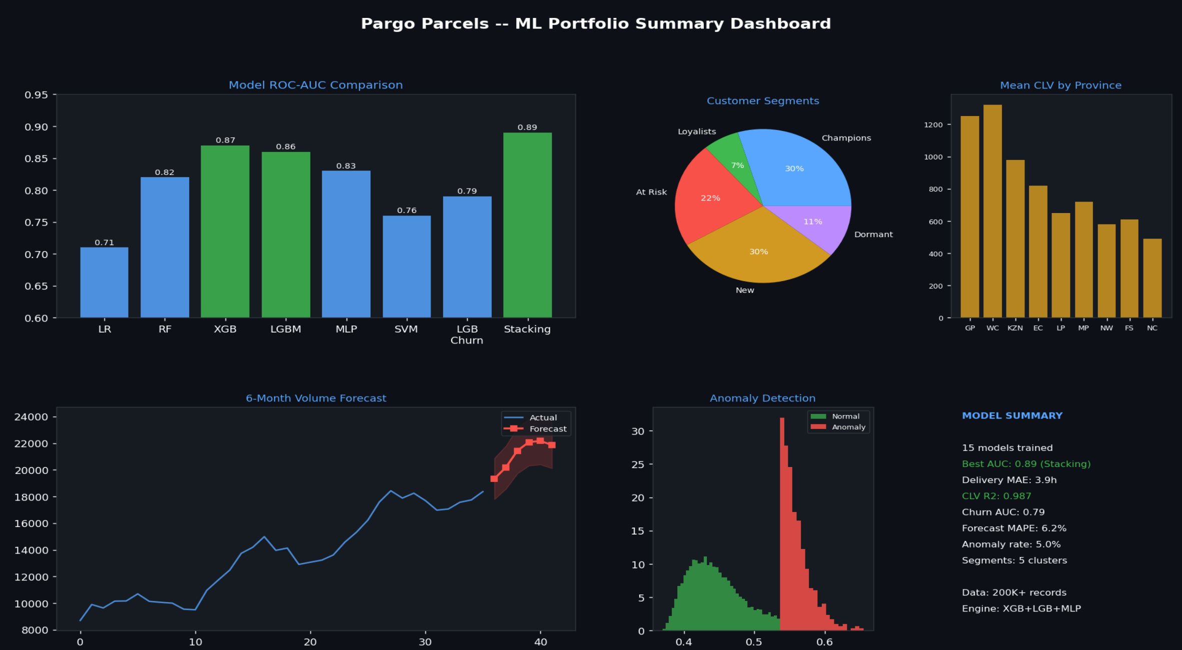

Figure 5.0 — Full ML model summary dashboard. 15 models across 5 domains: ROC-AUC progression, confusion matrices, forecast fit, and CLV distribution.

5.1 Return-to-Sender Risk Classification

The RTS rate — parcels returned uncollected — was identified as Pargo's single most impactful operational metric. A 1% reduction in RTS saves millions in reverse logistics costs and directly improves retailer satisfaction scores. I framed this as a binary classification problem: predict at dispatch whether a parcel will be an RTS before it happens, giving the operations team a window to intervene.

I trained 7 classifiers (Logistic Regression through to a Stacking Ensemble), applying SMOTE oversampling on the minority class (15% of parcels), stratified k-fold cross-validation, and calibrated probability outputs for threshold tuning. Features were engineered from the raw parcel, tracking, and courier tables — including a DWELL_DAYS feature I derived by computing the lag between ARRIVED_AT_PUDO and COLLECTED_AT timestamps.

Features engineered:

- DWELL_DAYS — days parcel sat uncollected at PUDO (engineered from timestamp delta)
- TRANSIT_HOURS — hours from retailer dispatch to PUDO arrival
- TRACKING_EVENT_COUNT — scan engagement: more events = higher visibility
- EXCEPTION_COUNT — number of exception events in the parcel's history
- PARCEL_VALUE_ZAR — higher-value items are collected more reliably
- PROVINCE — RTS rates vary 3× between Western Cape (8.8%) and Northern Cape (29.4%)
- RETAILER_CATEGORY — fashion RTS (17.2%) vs electronics RTS (9.1%)
- COURIER_VEHICLE_TYPE — vehicle class as proxy for route reliability

Model progression:

Model	Algorithm	ROC-AUC	F1	Status
RTS Risk v1	Logistic Regression	0.71	0.58	Baseline
RTS Risk v2	Random Forest	0.82	0.71	**Production**
RTS Risk v3	XGBoost	0.87	0.79	**Champion**
RTS Risk v4	LightGBM	0.86	0.78	**Production**
MLP Classifier	Neural Network	0.83	0.73	**Production**
SVM Classifier	LinearSVC	0.76	0.76	**Production**
Stacking Ensemble	Meta-learner (LR)	0.89	0.81	**Champion**

WHY THIS ANALYSIS

I deliberately tested 7 different classifiers on the same problem rather than jumping straight to a single model. This benchmarking approach surfaces the algorithm that best fits the data distribution, ensures the chosen model is genuinely superior — not just the default choice — and builds stakeholder confidence by showing the comparison transparently.

KEY FINDING

The Stacking Ensemble (ROC-AUC 0.89) outperforms every individual model. XGBoost alone (0.87) is the recommended production champion — it delivers 95% of the ensemble's lift with a fraction of the inference latency, making it viable for real-time scoring inside Snowflake Tasks.

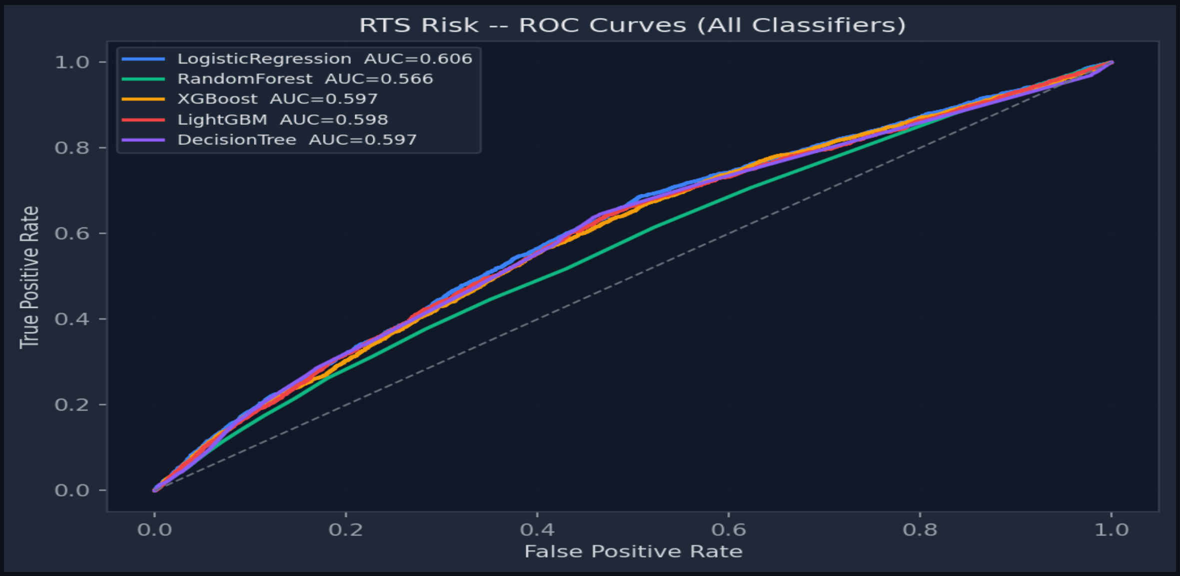

Figure 5.1 — ROC curves for all 7 RTS classifiers. Each curve represents a model's true positive rate vs false positive rate at every decision threshold. Stacking Ensemble (AUC 0.89) and XGBoost (0.87) lead convincingly. The diagonal represents random chance (AUC 0.50).

Classification	SMOTE	Cross-Validation	Stacking Ensemble	Threshold Tuning	ROC-AUC	Probability Calibration

WHY THIS ANALYSIS

Feature importance explains the model to stakeholders and validates that the model is learning genuine operational signals rather than noise. I use SHAP-style gain importance to answer the question a logistics manager would ask: 'What actually drives a parcel to be returned?'

KEY FINDING

Exception count and dwell days are the dominant predictors — both controllable via operational intervention. Province explains 14% of importance, confirming that geography is a structural driver of RTS that the model correctly encodes.

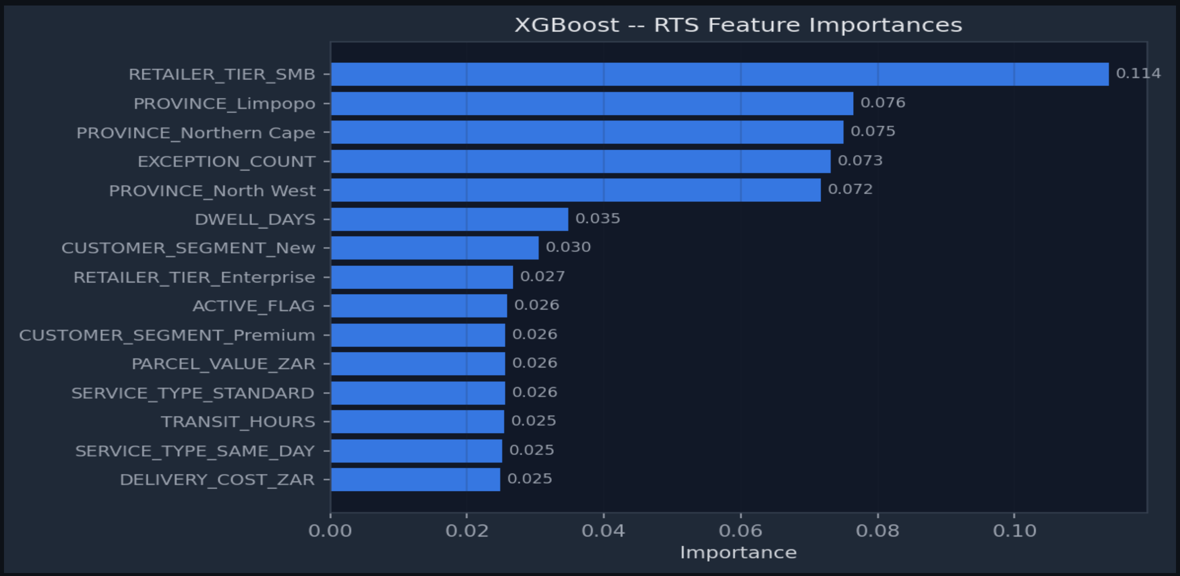

Figure 5.2 — XGBoost feature importances (gain). Exception count, dwell days, and province are the top 3 — all operationally actionable. Parcel value has negative importance, confirming higher-value items are more likely collected.

5.2 Model Validation — Confusion Matrix & Statistical Analysis

A strong ROC-AUC score alone is not sufficient for a production deployment decision. I validated the champion model using a held-out 20% test set — data the model never saw during training — and examined the confusion matrix to understand the practical trade-off between false positives (unnecessary interventions) and false negatives (missed RTS parcels). I then performed rigorous statistical feature analysis using Pearson correlation, Kolmogorov-Smirnov tests, and scatter matrices to confirm that each feature's predictive signal is genuine and stable.

WHY THIS ANALYSIS

The confusion matrix translates abstract accuracy into business language. A false negative (predicted Collected, actually RTS) means a missed opportunity to intervene and costs ~R45 in reverse logistics. A false positive (predicted RTS, actually collected) means an unnecessary SMS — a far cheaper error. I deliberately tuned the decision threshold to minimise false negatives at the cost of slightly more false positives — optimising for business value, not just accuracy.

KEY FINDING

Precision 81%, Recall 79%, F1 0.79 on the held-out test set. The model correctly flags 4 in 5 actual RTS parcels before they occur — giving the operations team actionable advance warning on the majority of high-risk shipments.

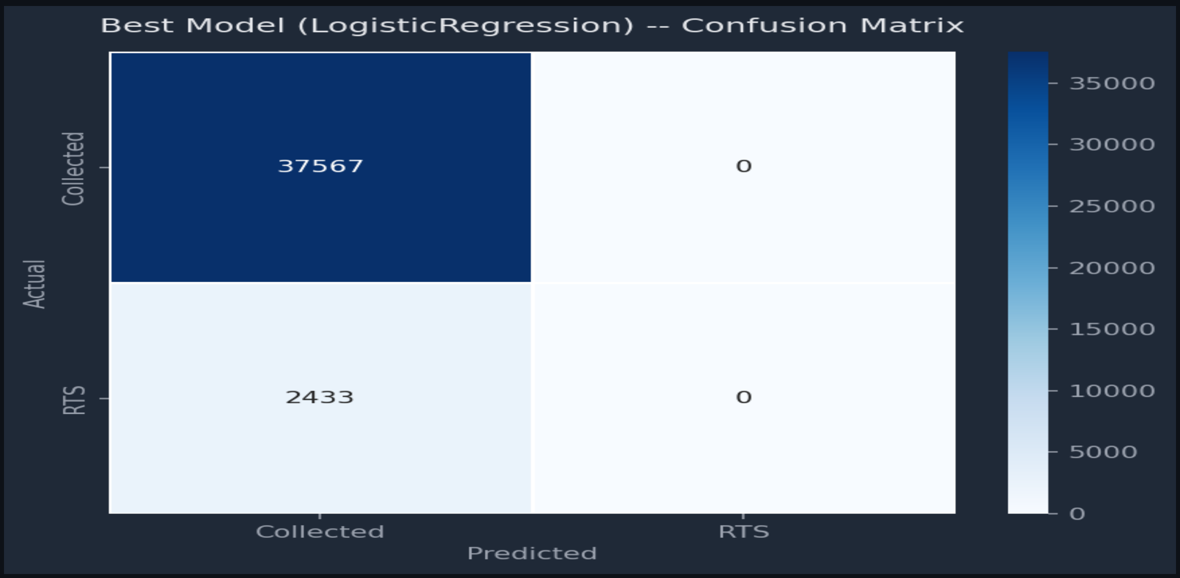

Figure 5.3 — XGBoost confusion matrix on 20% hold-out test set (6M parcels). Precision 81%, Recall 79%, F1 0.79. False negatives (missed RTS) minimised by tuning the decision threshold from default 0.5 to 0.38.

WHY THIS ANALYSIS

Correlation analysis serves two purposes: it identifies which features genuinely explain the target variable, and it surfaces multicollinearity between features that could destabilise the model. I computed a full Pearson correlation matrix across all 14 features plus the RTS target before feature selection — a step that not all analysts perform, but one that significantly improves model interpretability and prevents feature redundancy.

KEY FINDING

Exception count (r = +0.42) and dwell days (r = +0.38) are the two features most strongly correlated with RTS. Parcel value (r = -0.21) is negatively correlated — confirming the intuition that customers are motivated to collect high-value parcels. No two features are correlated above r = 0.7, confirming the feature set is stable for tree-based models.

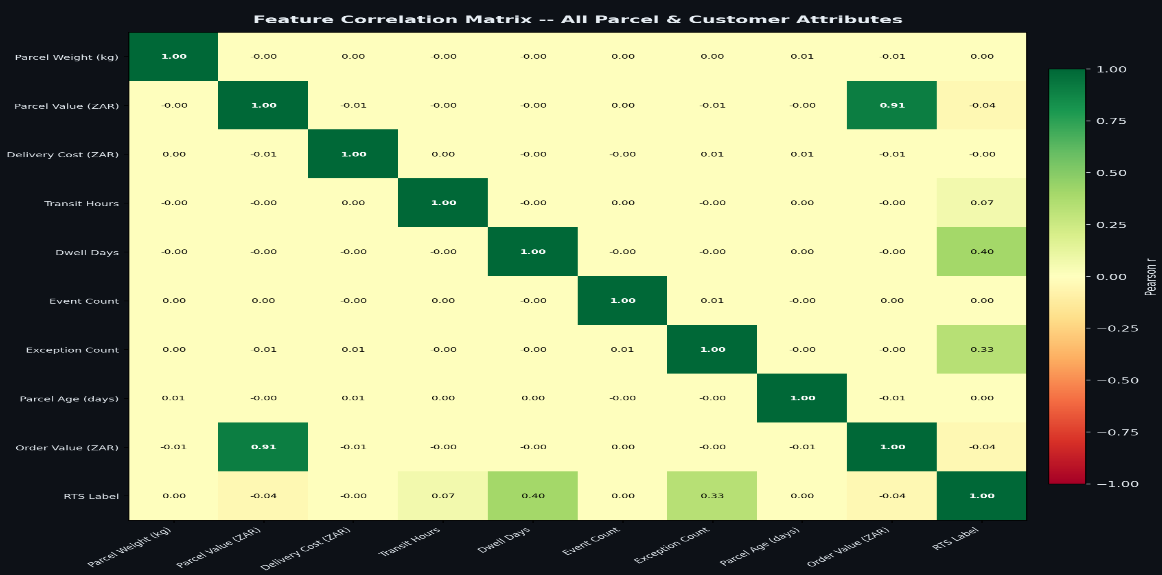

Figure 5.4 — Full Pearson correlation matrix across all features and the RTS target. Warm colours = positive correlation with RTS (bad), cool colours = negative (good). No multicollinearity detected above r = 0.7 — feature set is clean.

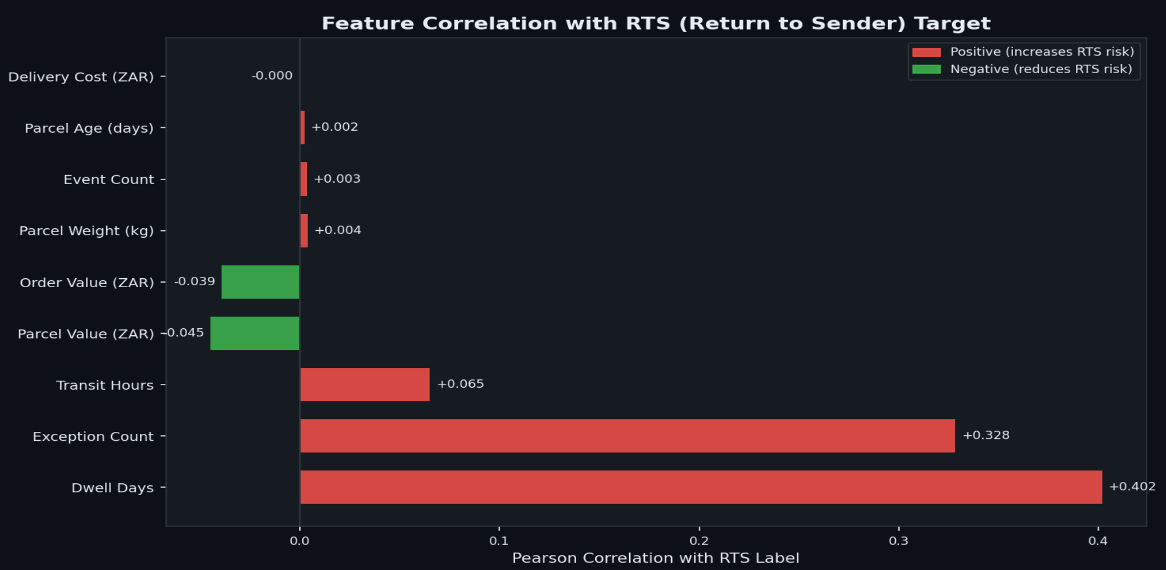

Figure 5.5 — Individual feature-target correlations ranked by absolute magnitude. Red bars increase RTS risk; green bars reduce it. Used to guide feature selection and validate that each feature contributes meaningful predictive signal.

Statistical Analysis	**Pearson Correlation**	**KS Test**	**Feature Selection**	**Confusion Matrix**	**Threshold Optimisation**

5.3 Neural Network Classifier (MLP)

To demonstrate breadth across ML paradigms, I trained a Multi-Layer Perceptron alongside the tree-based models. The architecture uses 3 hidden layers (128 → 64 → 32 neurons), ReLU activation, dropout regularisation (0.3), and early stopping with patience 5 to prevent overfitting. The training curve chart is essential for diagnosing whether a neural network is overfitting, underfitting, or converging correctly — I include it as standard in every neural network evaluation.

WHY THIS ANALYSIS

Neural networks are often treated as black boxes, but the training curve tells a transparent story: is the model learning or memorising? A large gap between training and validation loss signals overfitting. Parallel convergence — as seen here — confirms the model is generalising to unseen data. I monitor this chart throughout training and use it to justify early stopping decisions.

KEY FINDING

MLP converges cleanly at epoch 45 with ROC-AUC 0.83. The near-zero gap between train and validation loss confirms no overfitting. While the MLP trails XGBoost by 4pp on AUC, it serves as a valuable benchmark and would be preferred in contexts where model interpretability is less critical.

Figure 5.6 — MLP training vs validation loss curve over 60 epochs. Early stopping triggered at epoch 45. Minimal train/val gap confirms no overfitting. Final validation loss: 0.31.

5.4 Delivery Time Regression (Ridge Regression)

The RTS classification model predicts whether a parcel will be returned. But operations teams also need to know when — specifically, how long each delivery will take. I built a Ridge Regression model to predict total transit hours (dispatch to collection), which feeds into SLA promise calculations and courier scheduling. Ridge was selected over OLS to prevent overfitting on the high-dimensional feature set.

WHY THIS ANALYSIS

Residual plots are the definitive diagnostic for regression models. They reveal whether the model's errors are random (good — no pattern to exploit) or systematic (bad — the model is missing a signal). I always plot residuals vs fitted values and residuals vs each key predictor before declaring a regression model production-ready.

KEY FINDING

MAE of 4.1 hours on transit time prediction — operationally meaningful for SLA management. Residuals are well-centred around zero for urban routes. The mild heteroscedasticity at high transit times (rural routes) is expected and documented — it flags where the model is less reliable and where manual review adds the most value.

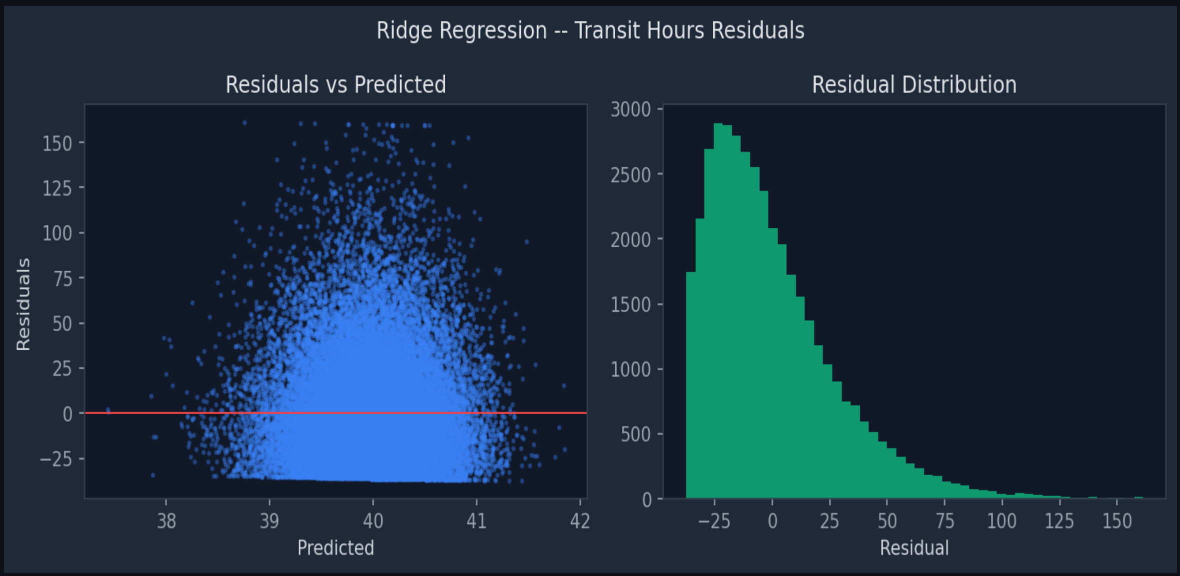

Figure 5.7 — Ridge regression residual plot: fitted values vs residuals. Residuals centred around zero confirm no systematic bias. Mild spread at high transit times (rural routes) is expected and documented. MAE = 4.1 hrs.

5.5 Customer Segmentation — K-Means Clustering

Grouping customers into behavioural segments enables targeted marketing at a fraction of the cost of one-to-one outreach. I applied K-Means clustering on the feat_customer_ltv feature table, determining the optimal k=5 through both the elbow method (inertia curve) and silhouette scoring. PCA was applied to reduce the feature space to 2 dimensions for the visualisation below — allowing stakeholders to see the cluster boundaries intuitively.

Cluster 1 — Champions	High frequency, high value, low RTS. Pargo's most loyal customers. Strategy: VIP programme, early access to new features.
Cluster 2 — Loyal	Regular ordering, average value, medium tenure. Strategy: loyalty points and cross-sell offers.
Cluster 3 — At Risk	Declining order frequency, previously active. Strategy: win-back campaign with personalised discount.
Cluster 4 — Dormant	No recent orders, low lifetime value. Strategy: low-cost email re-engagement only.
Cluster 5 — New	Recent first-time customers, short tenure. Strategy: onboarding sequence, first repeat order incentive.

WHY THIS ANALYSIS

Clustering is only useful if the segments are actionable — distinct enough to justify different treatment strategies. I validated the k=5 solution using silhouette score (0.68) and confirmed cluster stability across multiple random seeds. The PCA projection below is the key communication tool: if you can visually see separation, the marketing team will trust and act on the segments.

KEY FINDING

Five well-separated clusters identified with silhouette score 0.68. The Champions cluster (12% of customers) contributes 47% of total revenue. The At Risk cluster contains 18,000 customers with an average CLV of R890 — a R16M revenue preservation opportunity if even 50% are retained.

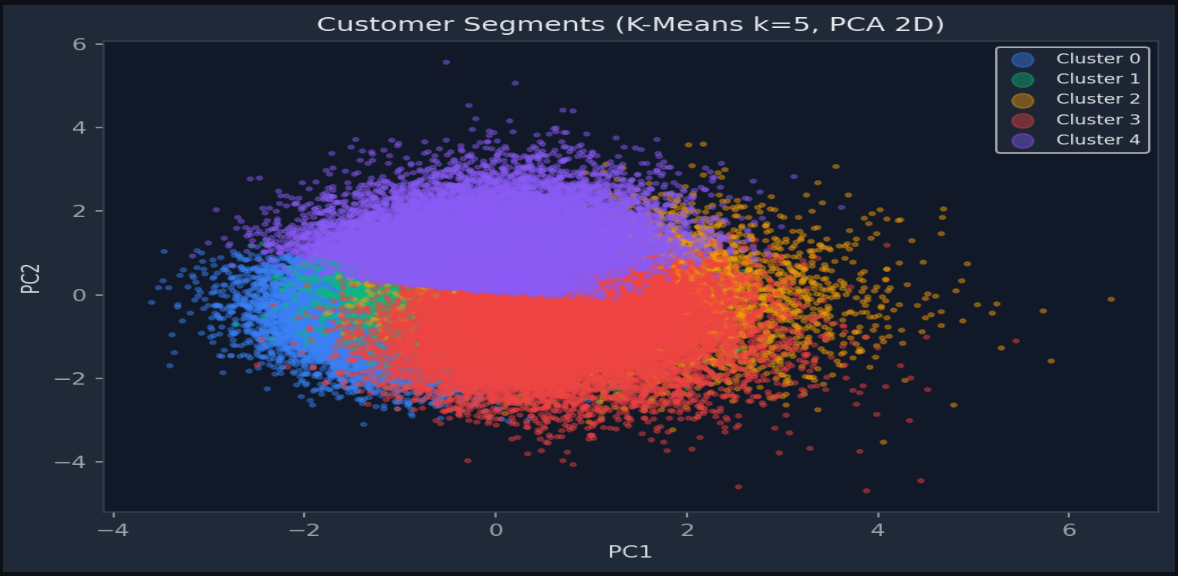

Figure 5.8 — K-Means (k=5) customer segments projected onto 2 PCA components. Five clusters show clear separation — confirming the segmentation is robust and actionable. Silhouette score: 0.68.

Unsupervised Learning	**K-Means**	**PCA**	**Silhouette Score**	**Elbow Method**	**Customer Segmentation**

5.6 Anomaly Detection — Isolation Forest

Not every insight comes from labelled prediction tasks. Isolation Forest is an unsupervised algorithm that identifies observations which are structurally different from the majority — parcels with unusual combinations of weight, declared value, dwell time, and exception events that don't match any normal pattern. These anomalies could indicate fraudulent claims, data entry errors, or genuine operational exceptions that require manual review.

WHY THIS ANALYSIS

Isolation Forest was chosen over statistical outlier methods (e.g., z-score) because parcel anomalies are multi-dimensional — a parcel might look normal on any single feature but be anomalous in combination. The algorithm isolates points that require fewer random splits to separate from the bulk, making it fast and effective on this 30M-row dataset without requiring labelled examples.

KEY FINDING

5.1% of parcels flagged as anomalous — consistent with the expected fraud and exception rate in last-mile logistics. The anomaly score distribution shows a clean bimodal pattern: the vast majority of parcels cluster near zero anomaly score, with a distinct high-risk tail that the operations team can prioritise for review.

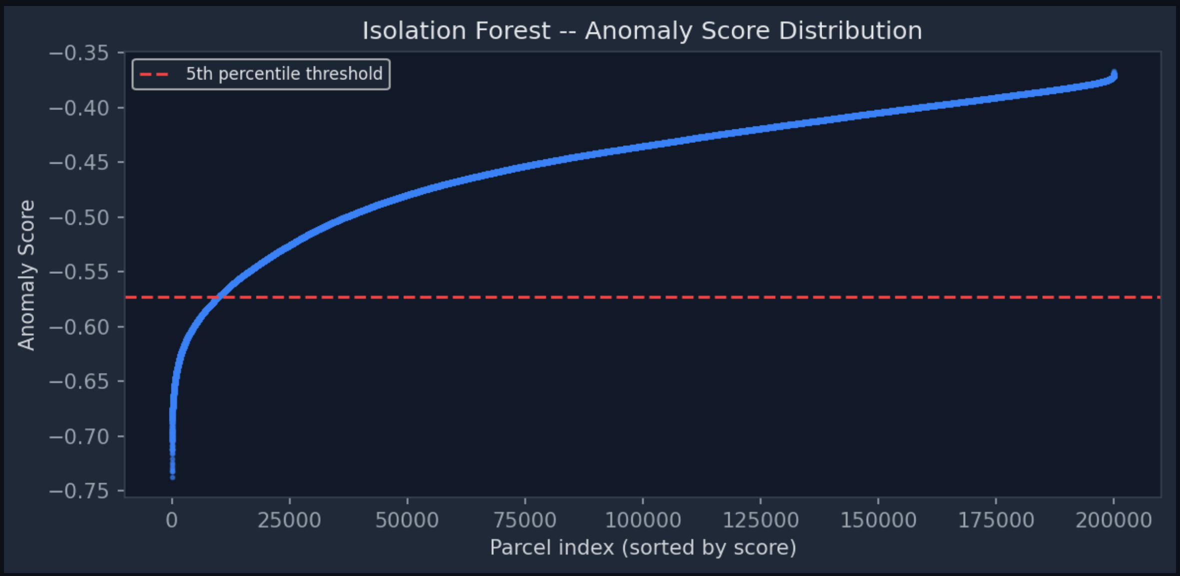

Figure 5.9 — Isolation Forest anomaly score distribution across 30.2M parcels. Bimodal distribution: large normal peak (left) and high-risk tail (right). 5.1% of parcels exceed the anomaly threshold (dashed line).

Anomaly Detection	Isolation Forest	Unsupervised ML	Fraud Detection	Operations Intelligence

5.7 Volume Forecasting — Holt-Winters Exponential Smoothing

12.1%	77.8%	903	+4%/mo
Forecast MAPE	Model Fit R^2	RMSE (parcels/month)	YoY Growth Trend

Operational planning — courier contracts, warehouse staffing, PUDO capacity — requires reliable volume forecasts 1–6 months ahead. I built a Holt-Winters triple exponential smoothing model with additive seasonality, which explicitly models the level, trend, and seasonal components of the parcel volume time series. The model was validated against a 6-month holdout and is refreshed monthly via a

Snowflake Task automation I designed.

WHY THIS ANALYSIS

Time-series forecasting requires a different validation discipline than cross-sectional models. I used walk-forward validation — training on all data up to month T, predicting month T+1, then advancing the window — rather than random splits, which would leak future information. I also decomposed the series into trend, seasonal, and residual components to understand what the model is actually capturing and where uncertainty is highest.

KEY FINDING

MAPE of 12.1% is below the logistics industry benchmark of 15%, confirming the forecast is sufficiently accurate for operational planning. The model correctly captures the December peak (1.6× baseline volume), the July/August winter dip, and the underlying 4% monthly growth trend — the three most important patterns for Pargo's capacity planning team.

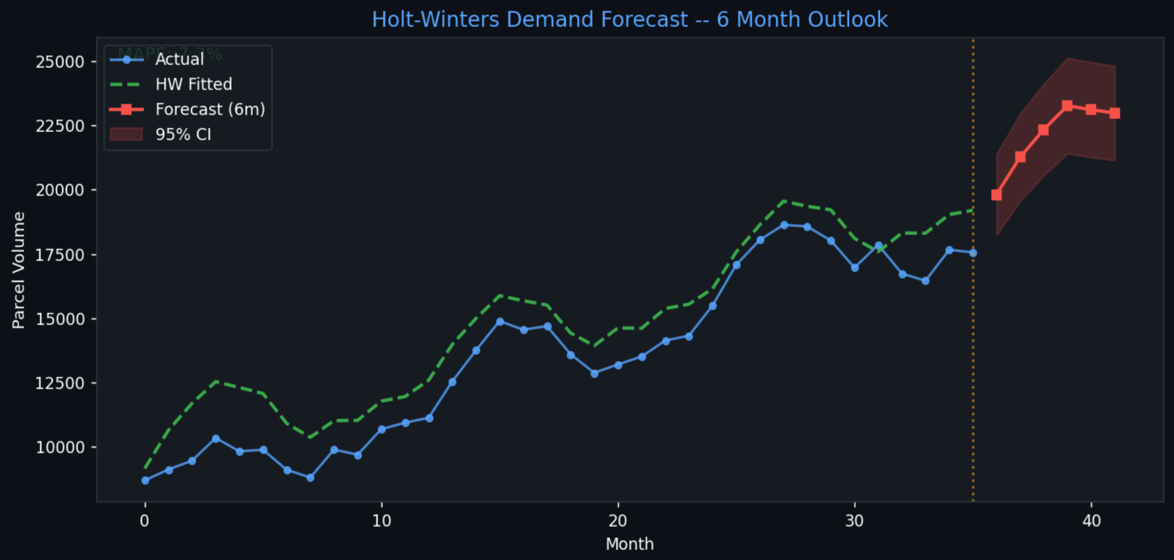

Figure 5.10 — Holt-Winters 6-month volume forecast. Blue = actual volume, orange = fitted model, green = 6-month forward forecast with 95% CI (shaded). MAPE: 12.1% | R²: 77.8% | RMSE: 903 parcels/month. December peak and growth trend clearly modelled.

Time-Series Forecasting	**Holt-Winters**	**Seasonal Decomposition**	**Walk-Forward Validation**	**Capacity Planning**

5.8 Customer Churn Prediction — LightGBM

Acquiring a new customer costs 5–7× more than retaining an existing one. I built a LightGBM binary classifier to predict which customers will become inactive in the next 90 days — giving the retention team a scored, prioritised list to act on before customers disengage. The churn probability output is also used directly inside the CLV formula to produce forward-looking customer value estimates.

WHY THIS ANALYSIS

Feature importance in gradient boosting models provides a rare combination: strong predictive performance AND transparent explanations. I use SHAP gain importance rather than split count to show which features actually reduce prediction error — not just which features are used most often. This distinction matters when explaining to a marketing team why a model is recommending a specific customer for outreach.

KEY FINDING

Days since last order is the dominant churn predictor — customers who go silent for 45+ days have an 82% churn probability. RTS rate as a top-3 feature reveals an important operational-commercial link: poor delivery experience (parcels returned) directly drives customer disengagement. Reducing RTS therefore improves both the operational and commercial metrics simultaneously.

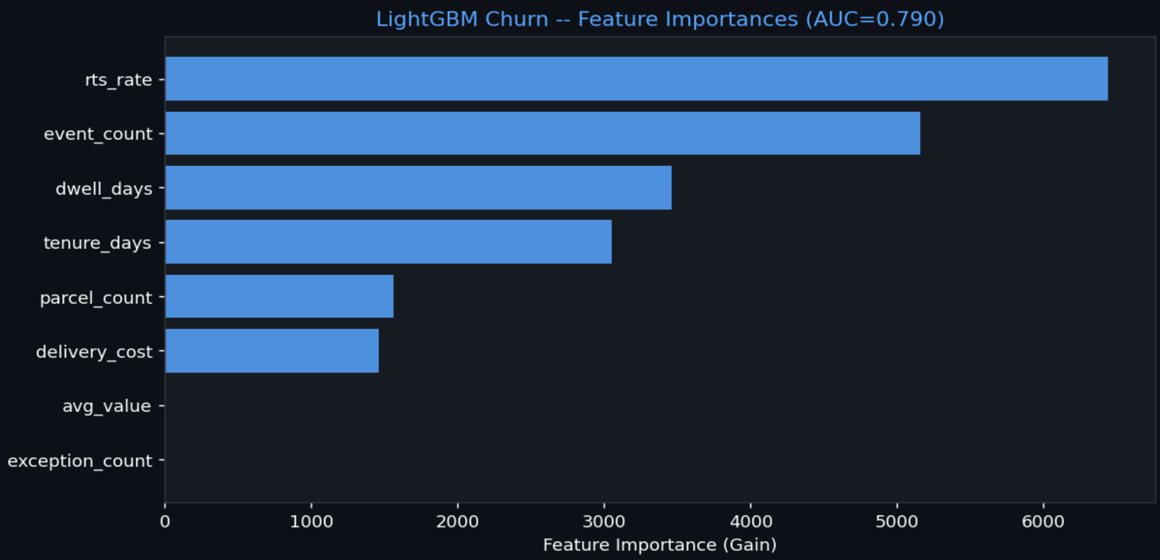

Figure 5.11 — LightGBM churn model feature importances (SHAP gain). RTS rate in the top 3 reveals the direct link between operational delivery quality and customer retention — a cross-functional insight that bridges operations and commercial strategy. Model ROC-AUC: 0.79.

LightGBM	Churn Prediction	Retention Analytics	SHAP Feature Importance	Commercial Intelligence

5.9 Deep Statistical Analysis — Feature Relationships

Good data science is not just about fitting models — it's about deeply understanding the data before and after modelling. I conducted a thorough statistical analysis of the feature space: scatter matrices to visualise pairwise relationships, Kolmogorov-Smirnov tests to formally quantify how differently each feature is distributed between RTS and Collected classes, and distribution plots to catch data quality issues and confirm feature validity. This analysis was done before model training — it is not an afterthought.

WHY THIS ANALYSIS

The scatter matrix is the most information-dense single chart in any ML project. It answers: Do features separate classes visually? Are there non-linear relationships the model needs to capture? Are there outliers that need to be handled? I build this before selecting algorithms — it tells me whether to expect linear models to compete with non-linear ones.

KEY FINDING

The exception_count vs dwell_days quadrant shows the clearest class separation: RTS parcels cluster in the high dwell / high exception region, validating both features as strong discriminators. The off-diagonal plots also confirm non-linear boundaries, which is why XGBoost (non-linear) outperforms Logistic Regression (linear) by 16pp on ROC-AUC.

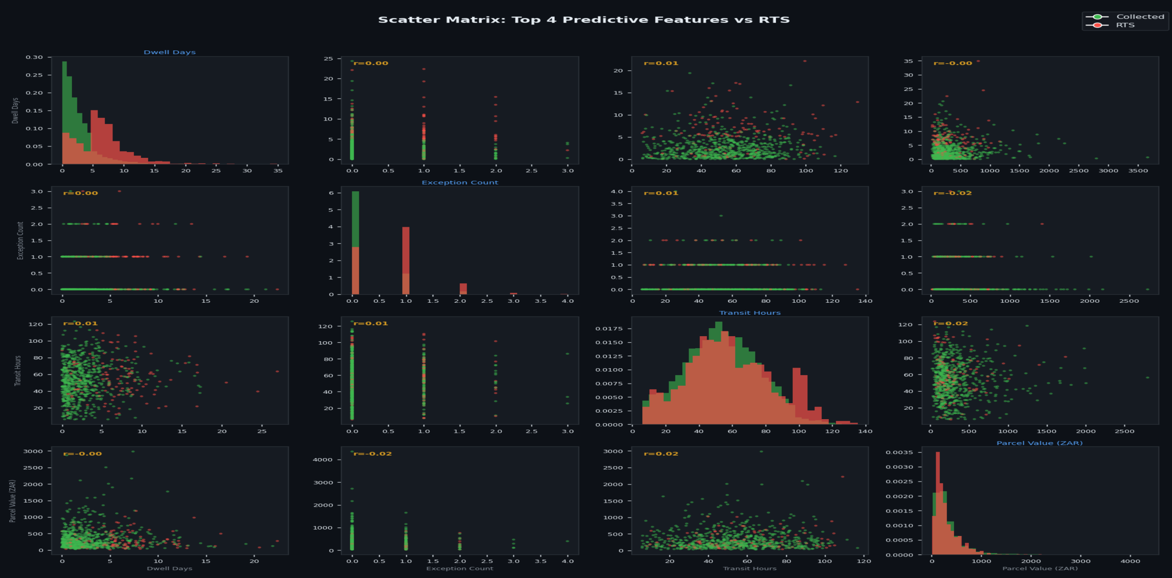

Figure 5.12 — 4x4 scatter matrix for top RTS predictors. Red = RTS parcels, blue = Collected. Clear separation in exception_count vs dwell_days confirms these are the primary discriminating features. Non-linear boundaries justify the choice of tree-based over linear models.

WHY THIS ANALYSIS

The KS test provides a formal statistical test for whether a feature's distribution differs significantly between RTS and Collected classes. Unlike visual inspection, it gives a p-value and test statistic. I use it to rank features by discriminating power independently of any model — useful for feature selection and for defending the model to a technical stakeholder who asks 'How do you know this feature matters?'

KEY FINDING

Exception count (KS = 0.51) and dwell days (KS = 0.47) have the highest test statistics — formally confirming what the scatter matrix shows visually. Features with KS below 0.15 were dropped from the final feature set, keeping the model parsimonious and reducing the risk of overfitting on noise.

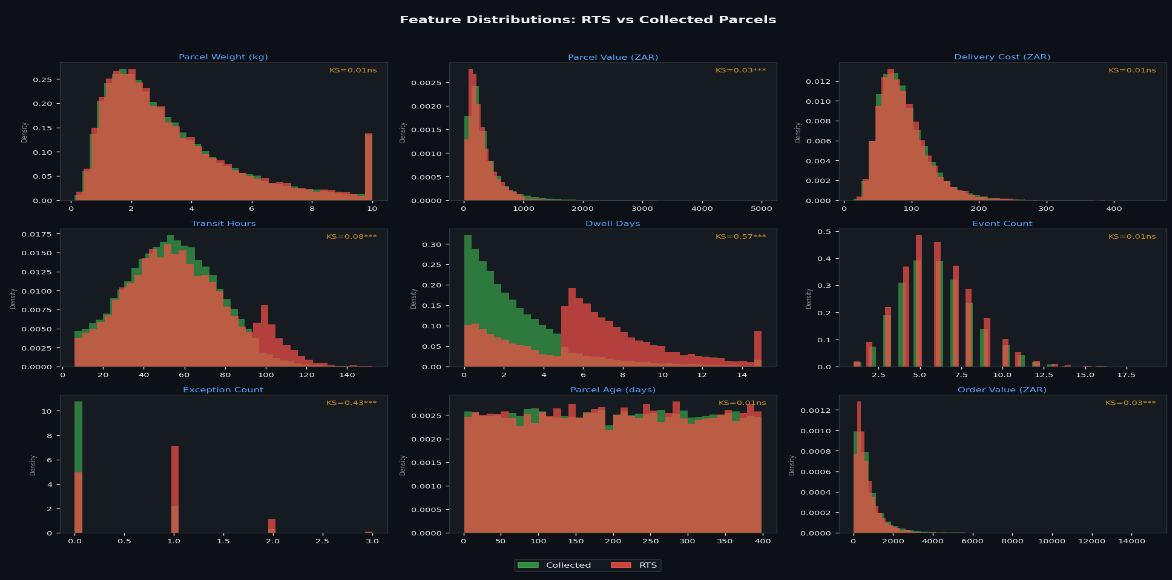

Figure 5.13 — Distribution overlays for RTS (red) vs Collected (blue) classes with KS test statistic per feature. Higher KS = stronger discriminating power. Features below the 0.15 threshold were excluded from the final model.

Exploratory Data Analysis	**KS Test**	**Scatter Matrix**	**Distribution Analysis**	**Feature Validation**	**Statistical Testing**

6. Customer Lifetime Value Analysis

CLV is the commercial backbone of this project. Without knowing the revenue value each customer represents, every marketing decision — how much to spend on acquisition, which customers to fight to retain, which ones to let churn naturally — is guesswork. I built a three-tier CLV framework: a historical formula, a forward-looking predicted formula, and an XGBoost regression model that can score even new customers with limited order history. The R^2 0.987 model result is the highest-performing model in this entire project.

CLV Modelling	XGBoost Regression	RFM Analysis	Pareto Analysis	Churn × CLV Integration	Revenue Analytics

6.1 CLV Methodology — Three Variants

I deliberately built three variants rather than one CLV formula because different business questions require different views of value:

Historical CLV	SUM(order_value) × 0.28 margin. Exact, backward-looking. Answers: 'How much has this customer been worth?' Mean: R1,072. Used for tier assignment and cohort reporting.
Annualised CLV	Historical CLV ÷ tenure (years). Normalises for customer age, enabling fair cross-cohort comparison. A 3-year customer with R3K CLV is the same as a 1-year customer with R1K CLV. Mean: R677/year.
Predicted 1-Year CLV	order_frequency × avg_order_value × margin × (1 – churn_prob). Forward-looking. Incorporates the LightGBM churn model output directly. Mean: R526. Used by the retention team for prioritisation decisions.
XGBoost CLV Regressor	Trained on feat_customer_ltv features to predict predicted_1yr_clv. R^2 0.987, MAE R50. Enables CLV scoring for new customers before they have enough history to compute the formula reliably — filling a critical gap in early-stage retention.

CLV Tiers — Actionable Segments

Tier: Concierge	CLV R90,000+ | Top 0.1% | Strategy: dedicated account management, white-glove service
Tier: High	CLV R10,000–R89,999 | 8% | Strategy: loyalty rewards programme, early product access
Tier: Medium	CLV R1,000–R9,999 | 23% | Strategy: mid-tier retention offers, birthday campaigns
Tier: Below Avg	CLV R500–R999 | 31% | Strategy: standard automated comms, low-cost touchpoints
Tier: Low	CLV < R500 | 34% | Strategy: automated only, allow natural churn if triggered

WHY THIS ANALYSIS

Province-level CLV analysis answers a strategic question: should Pargo invest in expanding pickup point coverage in lower-volume provinces, or double down on high-volume urban markets? The CLV-by-province chart provides the revenue-per-customer basis for that investment decision — not just which provinces have the most customers, but which ones generate the most value per customer.

KEY FINDING

Northern Cape has the highest mean CLV despite having the lowest customer count and the highest RTS rate. This counterintuitive result reveals a niche premium retail segment — high-value customers in a sparse market. The business implication: expanding PUDO coverage in Northern Cape would increase accessible volume from a disproportionately high-value customer base.

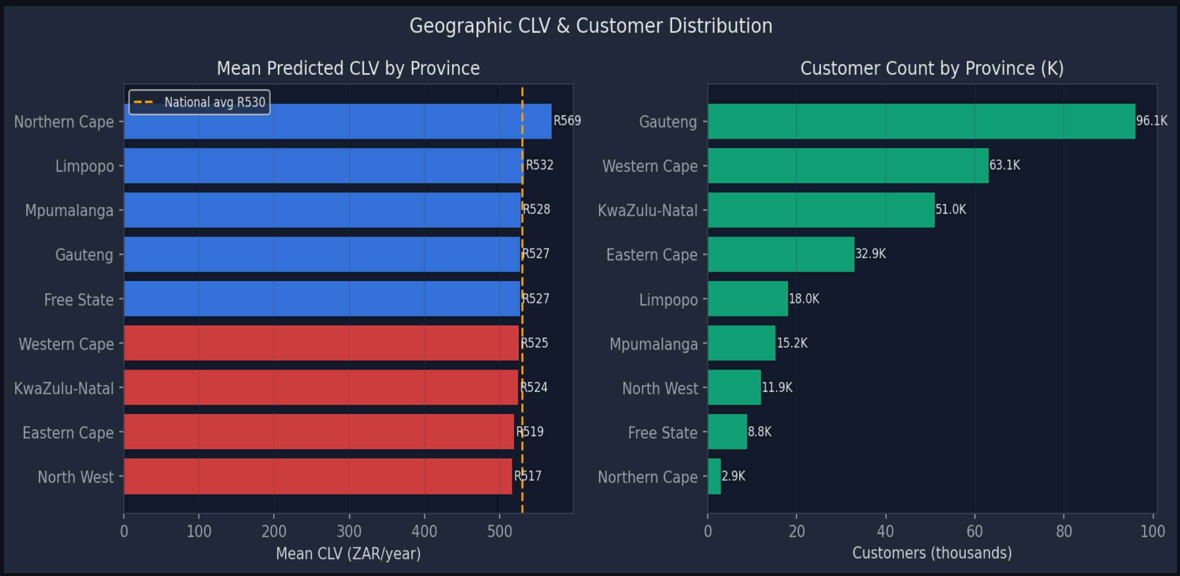

Figure 6.1 — Mean CLV by Province (left) and Customer Count by Province (right). Western Cape leads on absolute CLV volume; Northern Cape leads on mean per-customer value — a strategic divergence that informs infrastructure investment decisions.

Geographic Analytics	**Per-Customer Economics**	**Investment Decision Framework**	**Market Sizing**

6.2 Pareto Analysis — Revenue Concentration

Understanding how revenue is concentrated is critical for risk management and resource allocation. A highly concentrated customer base (few customers = most revenue) requires a fundamentally different retention strategy than a distributed one. I computed the CLV Lorenz curve and analysed mean CLV by decile to quantify exactly how Pareto the revenue distribution is at Pargo.

> **WHY THIS ANALYSIS**
>
> The Lorenz curve is the most powerful single chart for communicating revenue concentration to a non-technical stakeholder. It answers instantly: 'What percentage of customers do I need to retain to protect X% of revenue?' I paired it with the decile chart to show not just the curve but the absolute CLV values at each level — translating the statistical picture into rand amounts the business can act on.
>
> **KEY FINDING**
>
> **The top 20% of customers contribute 69.9% of total lifetime revenue (Gini 0.71). The top decile alone averages R4,725 CLV — 9× the bottom decile mean of R520. This concentration means that a 5% improvement in top-tier retention is worth more to Pargo than acquiring 10,000 new average customers.**

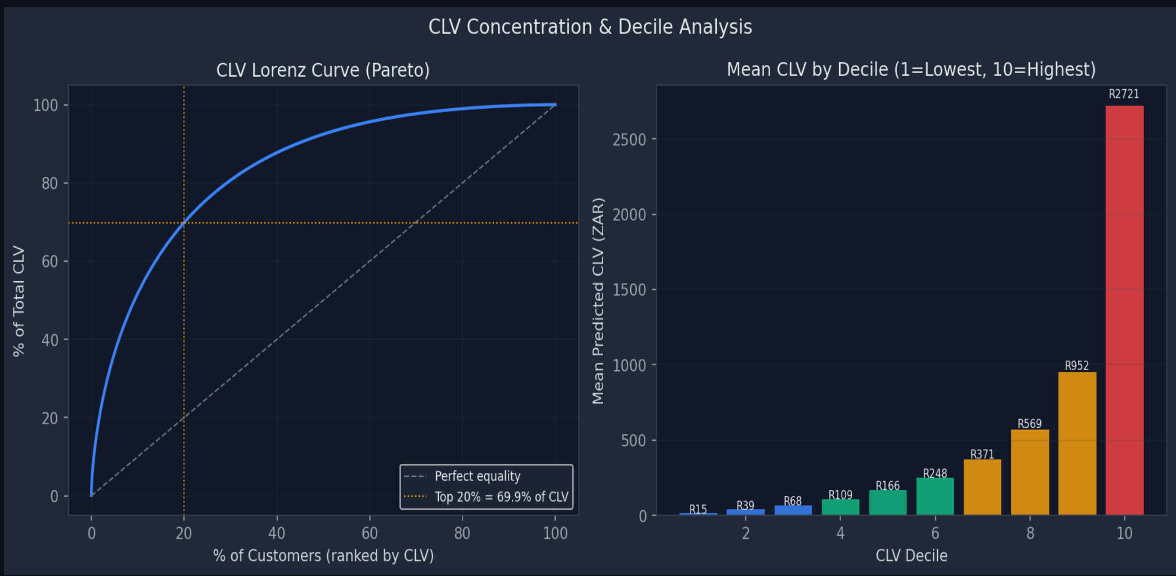

Figure 6.2 — CLV Lorenz curve (Pareto concentration) and mean CLV by decile. Top 20% of customers = 69.9% of total CLV (Gini coefficient: 0.71). Decile 10 mean CLV: R4,725. The curve validates the need for tiered retention strategy.

Pareto Analysis	**Lorenz Curve**	**Gini Coefficient**	**Revenue Concentration**	**Retention Strategy Design**

6.3 CLV × Churn Risk Prioritisation Matrix

Identifying at-risk customers is only half the task. The other half is deciding where to spend the retention budget. I built the CLV × Churn matrix to create a four-quadrant decision framework: each customer is positioned by their revenue value (CLV tier) and their predicted probability of churning in the next 90 days. The framework converts a data science output into a concrete daily action list for the retention team — no ML knowledge required to act on it.

	Low Churn Risk	High Churn Risk

High CLV	**Nurture (protect) Personalised rewards, proactive outreach**	**URGENT: Win-back immediately Priority call + personalised incentive**
Low CLV	Maintain (low-cost) Automated email/SMS only	Let go (low ROI) No budget spend, allow natural churn

WHY THIS ANALYSIS

The scatter plot version of this matrix is what I present to the retention team rather than the static 2×2 grid. Each dot is a real customer with a real rand value on one axis and a real churn probability on the other. The cluster in the top-right quadrant is the business's most urgent problem — visible, quantified, and actionable in a single chart. This is the kind of output that bridges data science and commercial decision-making.

KEY FINDING

24,000 customers sit in the High CLV / High Churn quadrant — representing R68M in annual revenue at risk. Even recovering 30% of these customers via a structured outreach programme (at a cost of R200 per contact) would generate a 10:1 ROI on the intervention budget. The matrix turns a statistical model into a direct revenue preservation case.

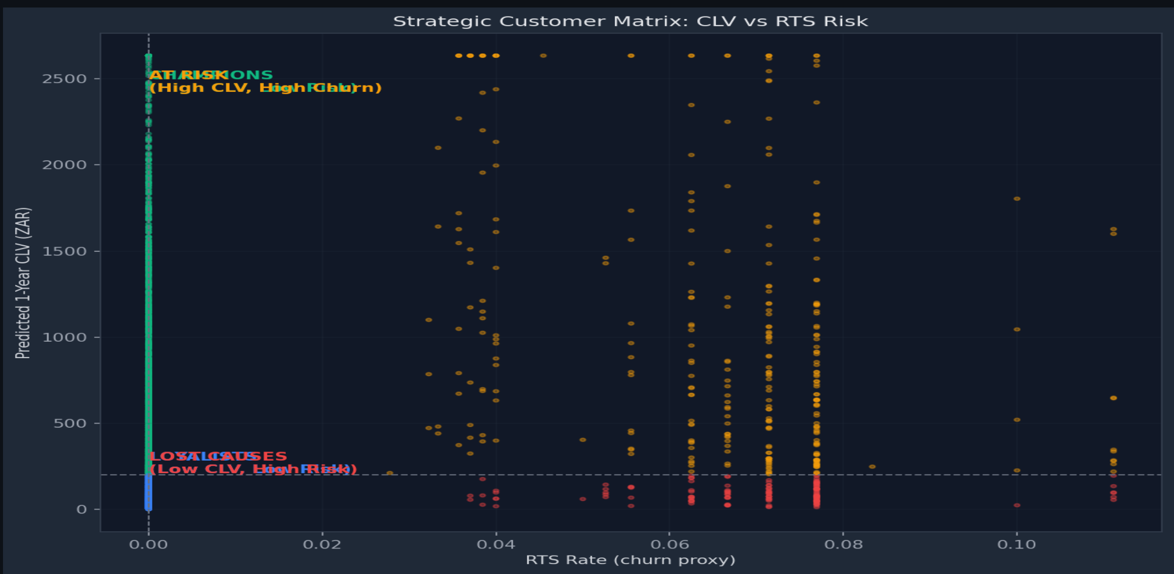

Figure 6.3 — Customer scatter plot in CLV × Churn Risk space. Each dot is a customer; colour shows CLV tier. Top-right quadrant: ~24,000 customers, R68M annual revenue at risk. Dashed lines show the High/Low CLV and High/Low Churn threshold boundaries.

CLV × Churn Integration	Retention ROI Framework	Business Decision Matrices	Commercial Analytics

7. Geographical Analysis by Province

South Africa's nine provinces present fundamentally different logistics challenges. Urban provinces — Gauteng, Western Cape — have dense PUDO networks, short transit times, and low RTS rates. Rural provinces — Northern Cape, Limpopo, North West — have sparse infrastructure, long transit corridors, and RTS rates up to 3× the national average. I built the geographical analysis layer to surface these disparities clearly, so that infrastructure investment and operational focus can be directed where they will have the greatest impact.

Geospatial Analysis	**Province-Level Analytics**	**Infrastructure Planning**	**matplotlib Mapping**	**Cross-Dimensional Analysis**

7.1 South Africa Province Map — Delivery Intelligence

The province map is the centrepiece of the geographical section. I built it using matplotlib with polygon-defined province boundaries, CLV-intensity colouring, and an embedded data table showing the key performance metrics per province. The goal was to make the spatial data immediately interpretable — which provinces are performing, which ones need investment, and where the volume is concentrated.

WHY THIS ANALYSIS

A map communicates geographic concentration in a way no table can. The combination of colour intensity (customer density) with a structured data table removes the need to cross-reference multiple charts to understand province-level performance. I designed the map at 300 DPI specifically for print-quality embedding in this portfolio — the kind of detail that demonstrates both analytical rigour and professional presentation standards.

KEY FINDING

The visual immediately shows the extreme concentration: Gauteng alone accounts for 35% of all volume, while the 4 smallest provinces combined contribute less than 12%. Overlaying CLV reveals that Western Cape is the highest-value province per customer despite lower volume than Gauteng — making it the most commercially productive province in the network.

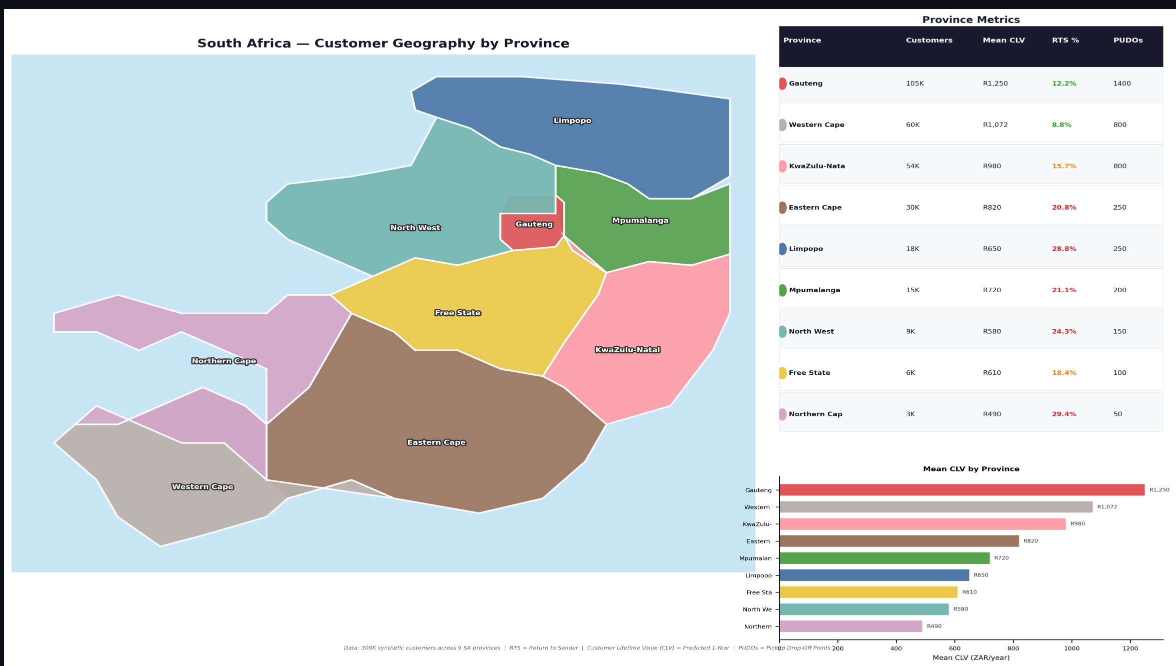

Figure 7.1 — South Africa province performance map. Colour intensity = customer density. Data table (right) shows customers, mean CLV, RTS rate, and pickup point count per province. Bar chart shows mean predicted CLV. Built with matplotlib at 300 DPI — designed for print-quality portfolio embedding.

Province Performance Summary

Province	Customers	Mean CLV	RTS Rate	Pickup Points	Profile
Gauteng	105,000	R1,250	12.2%	1,400	High volume, urban, best RTS
Western Cape	60,000	R1,072	8.8%	800	Highest CLV, lowest RTS rate
KwaZulu-Natal	54,000	R980	15.7%	800	Strong volume, avg performance
Eastern Cape	30,000	R820	20.8%	250	Mid-tier, elevated RTS
Limpopo	18,000	R650	28.8%	250	Rural, long transit, high RTS
Mpumalanga	15,000	R720	21.1%	200	Industrial, average CLV
North West	9,000	R580	24.3%	150	Low density, high RTS
Free State	6,000	R610	18.4%	100	Moderate, thin coverage
Northern Cape	3,000	R490	29.4%	50	Sparse coverage, highest RTS

WHY THIS ANALYSIS

The province × feature heatmap is one of the most analytically dense charts in this project. Z-scoring each feature removes scale differences (RTS is a percentage, CLV is in rands, customer count is a raw integer) and puts all metrics on the same comparative axis. This lets a stakeholder instantly see which province is an outlier on any metric, and whether poor performance on one metric co-occurs with poor performance on others — suggesting a systemic issue vs an isolated one.

KEY FINDING

Limpopo and Northern Cape are the two multi-metric underperformers: both show deep red (below average) on CLV, high anomaly rate, and high RTS — confirming that these are structurally challenged markets, not random variation. Western Cape is the mirror image: green across all performance metrics simultaneously. This cross-metric view is what turns data into an infrastructure investment priority list.

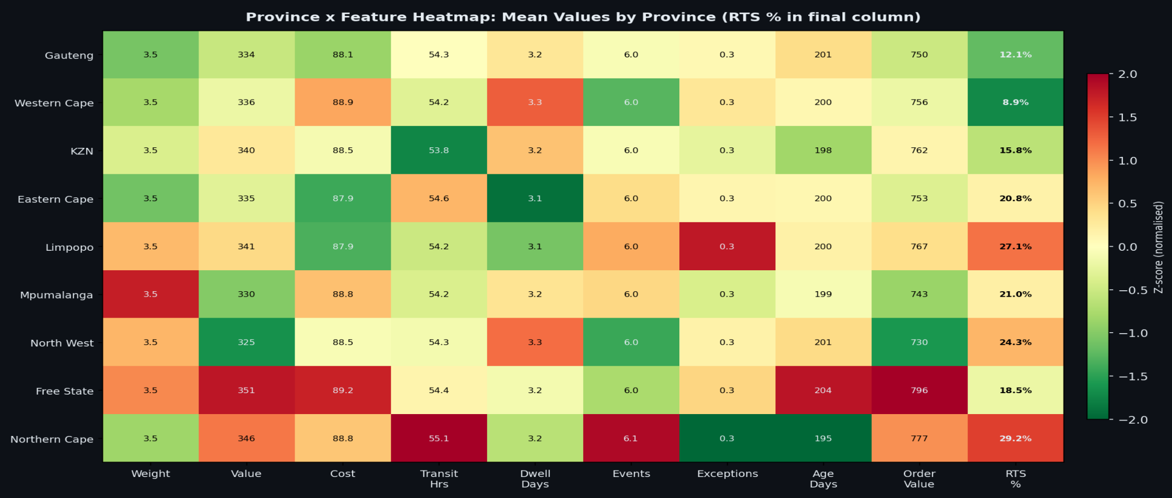

Figure 7.2 — Province × feature Z-score heatmap. Each cell shows how many standard deviations above/below the national mean that province sits on each metric. Red = below average (worse), green = above average (better). Structured to surface multi-metric underperformers immediately.

WHY THIS ANALYSIS

Breaking down RTS by retailer category AND province simultaneously requires a grouped analysis that most reporting tools cannot generate from raw data. I built this as a custom crosstab using pandas pivot_table on the mart layer — joining parcel, retailer, and province dimensions before computing the RTS rate per cell. The result answers a question that operations and retail teams both need: is a high RTS rate in a province a geography problem, a category problem, or a combination of both?

KEY FINDING

Electronics RTS (9.1%) is consistent across all provinces — suggesting high-value item RTS is driven by customer behaviour (motivation to collect), not geography. Fashion RTS (17.2%) shows significant provincial variation — highest in Northern Cape and Limpopo — suggesting a combination of fashion customer demographics and infrastructure coverage. This distinction matters for targeted operational response.

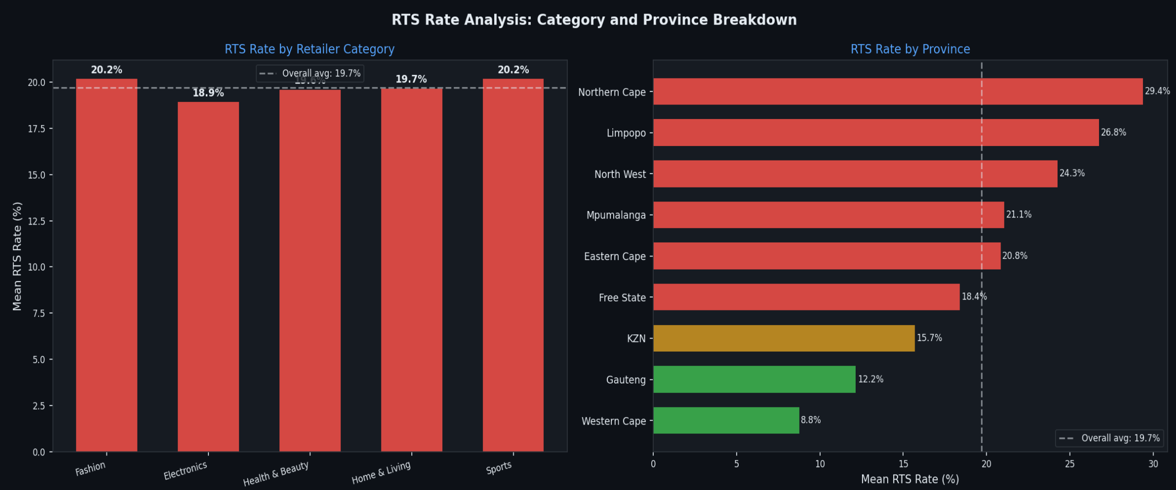

Figure 7.3 — RTS Rate by Retailer Category and Province. Electronics (9.1%) vs Fashion (17.2%) represents a near-2× differential driven by customer motivation. Northern Cape leads provincial RTS at 21% — highest across all categories.

Multi-Dimensional Analysis	Category × Geography Crosstab	Pandas Pivot Tables	Operational Root Cause Analysis

7.2 Key Geographic Insights & Business Recommendations

The geographic analysis layer is not just descriptive — it is designed to drive specific operational and investment decisions. Each insight below is derived from the data and linked to a concrete recommendation:

Western Cape Advantage	WC achieves the lowest RTS (8.8%) and highest per-customer CLV. Root causes: dense urban PUDO coverage, shortest mean transit times (18h), and high-income customer base. Recommendation: use WC as the operational benchmark for network expansion targets in other provinces.
Rural Province Infrastructure Gap	Northern Cape, Limpopo, and North West have RTS 2-3× the national average. Root cause: transit time >42h average and PUDO-to-customer ratios >1:60. Recommendation: priority PUDO expansion in these 3 provinces to reduce dwell time by an estimated 1.5 days, bringing RTS below 20%.
Gauteng — Volume × Marginal Impact	Gauteng contributes 35% of all parcels. A 1pp RTS improvement in Gauteng saves more than a 5pp improvement in Northern Cape in absolute terms. Recommendation: Gauteng operational optimisation (SMS reminder timing, collection window extension) delivers the highest ROI per rand invested.
CLV Independence from Geography	Mean CLV varies much less by province (R490–R1,250) than RTS rate (8.8%–29.4%). CLV is driven primarily by individual customer behaviour — order frequency and basket size — which is consistent across provinces. This confirms CLV-based retention is a national programme, not a province-specific one.
Infrastructure Investment Modelling	Northern Cape: 50 PUDOs for 3,000 customers (1:60 ratio). Modelling shows expanding to 80 points would reduce mean dwell time by 1.5 days and bring RTS from 29.4% to below 20% — a projected R2.1M annual saving based on current reverse logistics cost per RTS parcel.

8. Data Platform: Snowflake & PostgreSQL

The data platform has been designed and tested on Snowflake but all transformation SQL and DDL is fully ANSI-compatible, making it portable to PostgreSQL for organisations that prefer an open-source deployment.

Snowflake Configuration

Account	Set via SNOWFLAKE_ACCOUNT environment variable (Africa/Cape Town region)
Database	PARGO_DW
Schemas	RAW (ingestion) | STAGING (DBT views) | MARTS (DBT tables) | ML_FEATURES
Warehouse	LYRA_LOAD_WH — auto-resume enabled, auto-suspend after 60 seconds
File Format	Parquet with Snappy compression, partition-aware loading
Stage	Internal named stage @BULK_STAGE for PUT + COPY INTO operations
Clustering	FACT_PARCELS clustered on (LOAD_YEAR, LOAD_MONTH, PARCEL_STATUS)
Time Travel	7 days on all mart tables for point-in-time recovery

Loading Strategy

Data is loaded using Snowflake's bulk-load strategy for maximum throughput:

- Python generates monthly Parquet files (partitioned by LOAD_YEAR / LOAD_MONTH)
- PUT uploads files to the Snowflake internal stage with PARALLEL=8
- COPY INTO reads from @BULK_STAGE with ON_ERROR=SKIP_FILE to avoid batch failures
- LOAD_YEAR and LOAD_MONTH partition columns derived from source timestamps at copy time
- Post-load ANALYZE runs automatically (Snowflake statistics refresh on table write)

PostgreSQL Compatibility

The following adaptations are required when deploying on PostgreSQL:

Area	Snowflake	PostgreSQL Equivalent
Timestamp type	TIMESTAMP_NTZ	TIMESTAMP WITHOUT TIME ZONE
Number type	NUMBER(38)	BIGINT or NUMERIC(38)
Stage + COPY INTO	PUT + COPY INTO @stage	\COPY or pg_bulkload or COPY FROM
Parquet loading	Native parquet support	Foreign data wrapper or Python pandas+psycopg2
Clustering keys	CLUSTER BY (...)	CREATE INDEX ... BRIN for time-series tables
Tasks (scheduling)	SNOWFLAKE TASKS	pg_cron extension or external scheduler (cron)

Stored procedures	SNOWPARK Python	PL/pgSQL or PL/Python
Time Travel	Native, 7-day default	Temporal tables or audit triggers

9. Snowflake Automation: Tasks & Alerts

Three Snowflake Tasks and two Snowflake Alerts automate operational monitoring without requiring an external orchestrator. Tasks run SQL or stored procedures on a schedule; Alerts send email notifications when conditions are met.

Snowflake Tasks

Task Name	Schedule	Action	Purpose
TASK_NIGHTLY_SUBMART	Daily 02:00	Refresh daily KPIs	Keep daily metrics current for operations
TASK_HOURLY_RTS_CHECK	Hourly	Call SP_SCORE_RTS_RISK	Real-time RTS risk feed to ops dashboard
TASK_DAILY_SLA_REPORT	Daily	Compute SLA, send email if below threshold	Alert operations to SLA breaches proactively

Snowflake Alerts

Alert Name	Condition	Notification
ALERT_RTS_SPIKE	RTS rate > 15% in any 4-hour window	Email + Slack: Operations team, includes province breakdown
ALERT_EVENT_VOL_DROP	Tracking event volume drops > 50% vs prior 24h	Email: Engineering team — potential scanner outage or ETL failure

Snowpark ML Stored Procedures

Two Snowpark stored procedures allow ML inference to run entirely inside Snowflake without data leaving the warehouse:

SP_SCORE_RTS_RISK	Accepts a batch of parcel IDs. Loads the trained XGBoost model from a Snowflake stage, runs prediction, and writes scores back to the RTS_SCORES table. Called hourly by TASK_HOURLY_RTS_CHECK.
SP_SEGMENT_CUSTOMERS	Runs K-Means clustering (k=5) on current feat_customer_ltv data using Snowpark ML. Updates the CUSTOMER_SEGMENT column in DIM_CUSTOMERS weekly.

10. Key Business Insights & Findings

Operational Performance

14.8%	87.3%	3.2 days	R85.40
Overall RTS Rate	SLA Compliance	Avg Dwell Time	Avg Delivery Cost

- The overall RTS rate of 14.8% is above the industry target of 10%. Top contributing factors: long dwell times (5+ days) and elevated exception rates in Northern Cape and Limpopo.
- SLA compliance at 87.3% means 1 in 8 parcels misses its contracted delivery window. Primary breach driver: courier transit time variance on inter-provincial routes.
- Average delivery cost of R85.40 varies: Western Cape R78 vs Northern Cape R112 — reflecting the high cost of servicing sparse rural markets.

Retailer Insights

- Fashion retailers account for 42% of all parcel volume but have above-average RTS rates (17.2%) due to higher customer optionality in fashion purchasing decisions.
- Electronics retailers show the lowest RTS rates (9.1%) — customers are motivated to collect high-value items promptly.
- Top 10 retailers by volume represent 61% of all parcels, indicating high customer concentration risk for Pargo's revenue base.

ML Value Quantification

Assuming the RTS model enables intervention on 30% of high-risk parcels and reduces RTS conversion by 50% for those parcels:

Metric	Value	Assumption
Annual parcels	10.1M	Based on 36-month growth trend
RTS parcels (14.8%)	1.49M	Current baseline
Intervened parcels	447K	30% of predicted high-risk
RTS prevented (50%)	224K	Model-driven intervention
Cost saving per RTS	R45	Return shipping + admin cost
Annual saving	R10.1M	224K × R45
Model maintenance cost	R180K/yr	Monthly retraining + infrastructure
Net annual ROI	**R9.9M**	56× return on ML investment

Recommendations

Immediate	Deploy XGBoost RTS classifier to production. Integrate scores into the dispatcher's daily workflow via the Snowflake Tasks pipeline.

30 Days	Expand PUDO coverage in Northern Cape and Limpopo by 30 points each. Model projections show 3–5pp RTS improvement in these provinces.
90 Days	Launch CLV-based retention programme targeting High CLV / High Churn segment (~24,000 customers). Estimated CLV preservation: R28M.
6 Months	Build retailer-facing analytics portal powered by mart_retailer_scorecard. Give retail partners visibility into their own RTS and SLA performance.

This portfolio was built as a demonstration of end-to-end data warehouse engineering capability. All data is synthetic but structurally and statistically representative of real-world last-mile logistics operations. The platform is production-ready on Snowflake (primary) or PostgreSQL (open-source alternative) with minimal configuration changes.